

Bookstore Sales and Inventory Analysis Using SQL

BASIC SQL QUERIES

1. Retrieve all books in the "Fiction" genre

```
SELECT *  
FROM Books  
WHERE Genre = 'Fiction';
```

2. Find books published after the year 1950

```
SELECT *  
FROM Books  
WHERE Published_Year > 1950;
```

3. List all customers from Canada

```
SELECT *  
FROM Customers  
WHERE Country = 'Canada';
```

4. Show orders placed in November 2023

```
SELECT *  
FROM Orders  
WHERE Order_Date BETWEEN '2023-11-01' AND '2023-11-30';
```

5. Retrieve the total stock of books available

```
SELECT SUM(Stock) AS Total_Stock  
FROM Books;
```

6. Find the details of the most expensive book

```
SELECT Title,  
Price  
FROM Books  
ORDER BY Price DESC LIMIT 1;
```

7. Show all customers who ordered more than 1 quantity of a book

```
SELECT *  
FROM Orders  
WHERE Quantity > 1;
```

8. Retrieve all orders where the total amount exceeds \$20

```
SELECT *  
FROM Orders  
WHERE Total_Amount > 20;
```

9. List all genres available in the Books table

```
SELECT DISTINCT Genre  
FROM Books;
```

10. Find the book with the lowest stock

```
SELECT *  
FROM Books  
ORDER BY Stock ASC  
LIMIT 1;
```

11. Calculate the total revenue generated from all orders

```
SELECT SUM(Total_Amount) AS Total_Revenue  
FROM Orders;
```

ADVANCED SQL QUERIES

1. Retrieve the total number of books sold for each genre

```
SELECT b.Genre,  
SUM(o.Quantity) AS Total_Books_Sold  
FROM Books b  
JOIN Orders o  
ON b.Book_ID = o.Book_ID  
GROUP BY b.Genre;
```

2. Find the average price of books in the "Fantasy" genre

```
SELECT AVG(Price) AS Avg_Price  
FROM Books  
WHERE Genre = 'Fantasy';
```

3. List customers who have placed at least 2 orders

```
SELECT c.Customer_ID,  
c.Name,  
COUNT(o.Order_ID) AS Orders_Placed  
FROM Customers c  
JOIN Orders o  
ON c.Customer_ID = o.Customer_ID  
GROUP BY c.Customer_ID,  
c.Name  
HAVING COUNT(o.Order_ID) >= 2;
```

4. Find the most frequently ordered book

```
SELECT Book_ID,  
COUNT(Order_ID) AS Times_Ordered  
FROM Orders  
GROUP BY Book_ID  
ORDER BY Times_Ordered DESC  
LIMIT 1;
```

5. Show the top 3 most expensive books in the "Fantasy" genre

```
SELECT Title,  
Price  
FROM Books  
WHERE Genre = 'Fantasy'  
ORDER BY Price DESC  
LIMIT 3;
```

6. Retrieve the total quantity of books sold by each author

```
SELECT b.Author,  
SUM(o.Quantity) AS Total_Sold  
FROM Books b  
JOIN Orders o  
ON b.Book_ID = o.Book_ID  
GROUP BY b.Author;
```

7. List the cities where customers spent over \$30

```
SELECT DISTINCT c.City  
FROM Customers c  
JOIN Orders o  
ON c.Customer_ID = o.Customer_ID  
WHERE o.Total_Amount > 30;
```

8. Find the customer who spent the most on orders

```
SELECT c.Customer_ID,  
c.Name,  
SUM(o.Total_Amount) AS Total_Spent  
FROM Customers c  
JOIN Orders o  
ON c.Customer_ID = o.Customer_ID  
GROUP BY c.Customer_ID,  
c.Name  
ORDER BY Total_Spent DESC  
LIMIT 1;
```

9. Calculate the stock remaining after fulfilling all orders

```
SELECT b.Book_ID,  
b.Title,  
b.Stock,  
COALESCE(SUM(o.Quantity), 0) AS Total_Quantity_Sold,
```

```
b.Stock - COALESCE(SUM(o.Quantity), 0) AS Remaining_Quantity
FROM Books b
LEFT JOIN Orders o
ON b.Book_ID = o.Book_ID
GROUP BY b.Book_ID,
b.Title,
b.Stock;
```